

Student Observer: \_\_\_\_\_

Period: **PERIOD 0 HONORS**

Date: \_\_\_\_\_

 **OBSERVER PROTOCOL:** As each classmate executes their live demonstration, record the setup and physical regime. In your defense notes, explain why the inertial mass behaved as observed and verify that **zero forward force** acted on it. Debunk any claims that "inertia pushed it."

DEMO 1

Demonstrator: \_\_\_\_\_

Codename: \_\_\_\_\_

[ ] Rest [ ] Motion [ ] Direction

Apparatus & Fast Impulse:

Mass, support surface, and how impulse force was applied:

.....

.....

Observed Behavior & Physics Defense:

What did mass do? How does this prove Newton's 1st Law ( $\Sigma F = 0$ )?

.....

.....

Critical Check: Was any unphysical forward force claimed? [ ] Yes (Error) [ ] No (Scientifically Accurate)

Defense Score: [ 1 ][ 2 ][ 3 ][ 4 ] / 4 pts

DEMO 2

Demonstrator: \_\_\_\_\_

Codename: \_\_\_\_\_

[ ] Rest [ ] Motion [ ] Direction

Apparatus & Fast Impulse:

Mass, support surface, and how impulse force was applied:

.....

.....

Observed Behavior & Physics Defense:

What did mass do? How does this prove Newton's 1st Law ( $\Sigma F = 0$ )?

.....

.....

Critical Check: Was any unphysical forward force claimed? [ ] Yes (Error) [ ] No (Scientifically Accurate)

Defense Score: [ 1 ][ 2 ][ 3 ][ 4 ] / 4 pts

DEMO 3

Demonstrator: \_\_\_\_\_

Codename: \_\_\_\_\_

[ ] Rest [ ] Motion [ ] Direction

Apparatus & Fast Impulse:

Mass, support surface, and how impulse force was applied:

.....

.....

Observed Behavior & Physics Defense:

What did mass do? How does this prove Newton's 1st Law ( $\Sigma F = 0$ )?

.....

.....

Critical Check: Was any unphysical forward force claimed? [ ] Yes (Error) [ ] No (Scientifically Accurate)

Defense Score: [ 1 ][ 2 ][ 3 ][ 4 ] / 4 pts

DEMO 4

Demonstrator: \_\_\_\_\_

Codename: \_\_\_\_\_

[ ] Rest [ ] Motion [ ] Direction

Apparatus & Fast Impulse:

Mass, support surface, and how impulse force was applied:

.....

.....

Observed Behavior & Physics Defense:

What did mass do? How does this prove Newton's 1st Law ( $\Sigma F = 0$ )?

.....

.....

Critical Check: Was any unphysical forward force claimed? [ ] Yes (Error) [ ] No (Scientifically Accurate)

Defense Score: [ 1 ][ 2 ][ 3 ][ 4 ] / 4 pts

DEMO 5

Demonstrator: \_\_\_\_\_

Codename: \_\_\_\_\_

[ ] Rest [ ] Motion [ ] Direction

Apparatus & Fast Impulse:

Mass, support surface, and how impulse force was applied:

Observed Behavior & Physics Defense:

What did mass do? How does this prove Newton's 1st Law ( $\Sigma F = 0$ )?

Critical Check: Was any unphysical forward force claimed? [ ] Yes (Error) [ ] No (Scientifically Accurate)

Defense Score: [ 1 ][ 2 ][ 3 ][ 4 ] / 4 pts

DEMO 6

Demonstrator: \_\_\_\_\_

Codename: \_\_\_\_\_

[ ] Rest [ ] Motion [ ] Direction

Apparatus & Fast Impulse:

Mass, support surface, and how impulse force was applied:

Observed Behavior & Physics Defense:

What did mass do? How does this prove Newton's 1st Law ( $\Sigma F = 0$ )?

Critical Check: Was any unphysical forward force claimed? [ ] Yes (Error) [ ] No (Scientifically Accurate)

Defense Score: [ 1 ][ 2 ][ 3 ][ 4 ] / 4 pts

DEMO 7

Demonstrator: \_\_\_\_\_

Codename: \_\_\_\_\_

[ ] Rest [ ] Motion [ ] Direction

Apparatus & Fast Impulse:

Mass, support surface, and how impulse force was applied:

Observed Behavior & Physics Defense:

What did mass do? How does this prove Newton's 1st Law ( $\Sigma F = 0$ )?

Critical Check: Was any unphysical forward force claimed? [ ] Yes (Error) [ ] No (Scientifically Accurate)

Defense Score: [ 1 ][ 2 ][ 3 ][ 4 ] / 4 pts

DEMO 8

Demonstrator: \_\_\_\_\_

Codename: \_\_\_\_\_

[ ] Rest [ ] Motion [ ] Direction

Apparatus & Fast Impulse:

Mass, support surface, and how impulse force was applied:

Observed Behavior & Physics Defense:

What did mass do? How does this prove Newton's 1st Law ( $\Sigma F = 0$ )?

Critical Check: Was any unphysical forward force claimed? [ ] Yes (Error) [ ] No (Scientifically Accurate)

Defense Score: [ 1 ][ 2 ][ 3 ][ 4 ] / 4 pts

 HONORS SYNTHESIS: BRIDGING INERTIA TO BALANCED FORCES ( $\Sigma F = 0$ )

1. Forensic Analysis of Confounding Forces:

In every demonstration observed today, whenever the mass stayed at rest or moved at constant velocity, what was the horizontal net force acting on that mass? Write the equation:  $\Sigma F_x =$  \_\_\_\_\_ N. What would occur if an unbalanced net force acted?

HONORS LOG DEFENSE SCORE:

• Observation Panels (16 pts)

• Synthesis & Equilibrium Bridge (4 pts)

TOTAL: \_\_\_\_\_ / 20 pts

Teacher Signature: \_\_\_\_\_